Lecture: New Trade Theory

Attend your tutorial next week

- There will be a quiz, but does *not* count towards your mark.

Recap from last week:

-Old Trade Theory can explain trade in dissimilar goods and tasks between dissimilar nations.

-HO model predicts income pc catch-up for developing countries, if they follow free trade and remain open to multinationals.

Recap from last week:

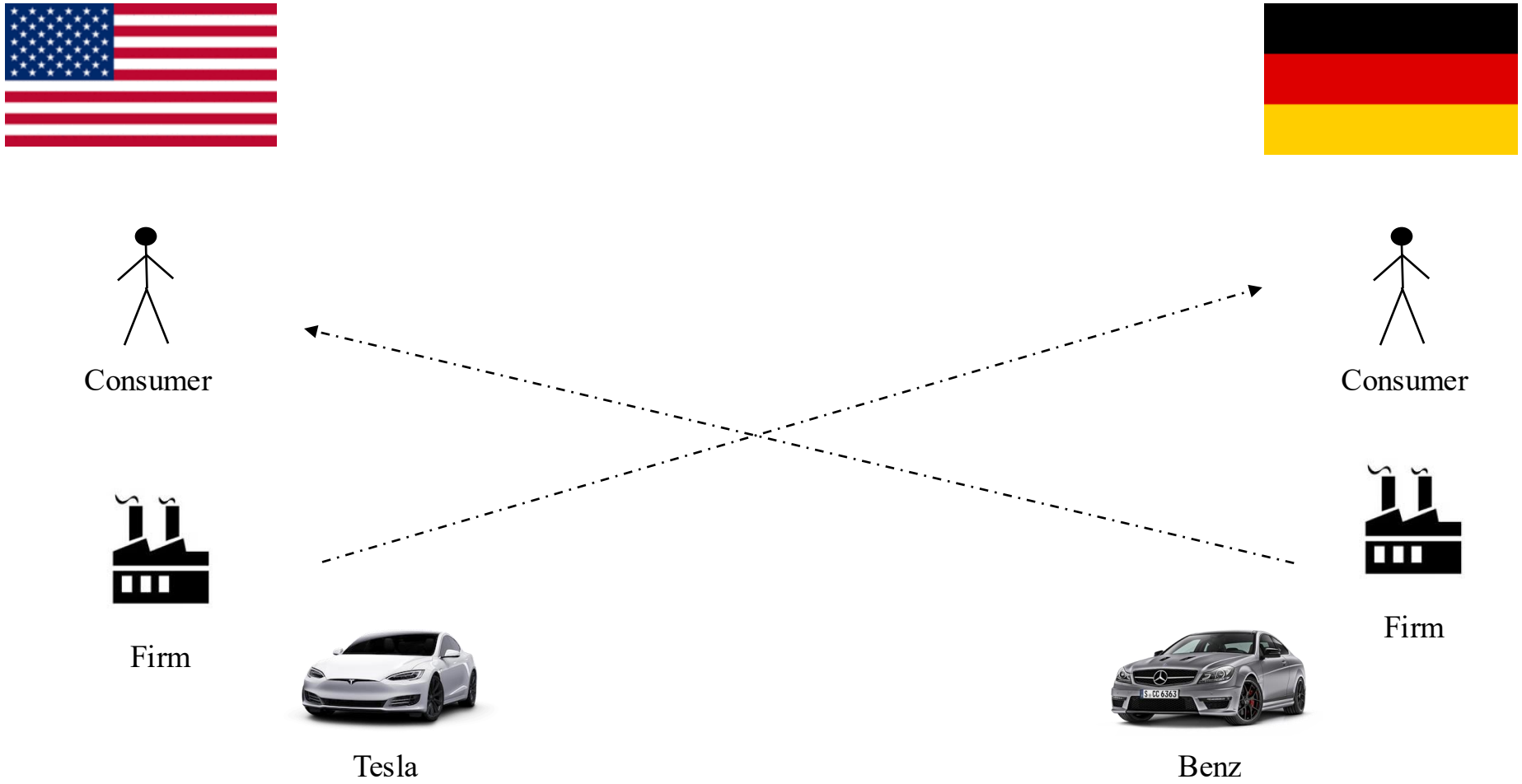
-Old Trade Theory can explain trade in dissimilar goods and tasks between dissimilar nations.

-HO model predicts income pc catch-up for developing countries, if they follow free trade and remain open to multinationals.

But there are two important things that Old trade theory can't explain

1- Why do countries trade in similar products?

1- Why do countries trade in similar products?



2- Why is outsourcing so concentrated in few developing countries?

2- Why is outsourcing so concentrated in few developing countries?

Selected specialised clusters in southern China

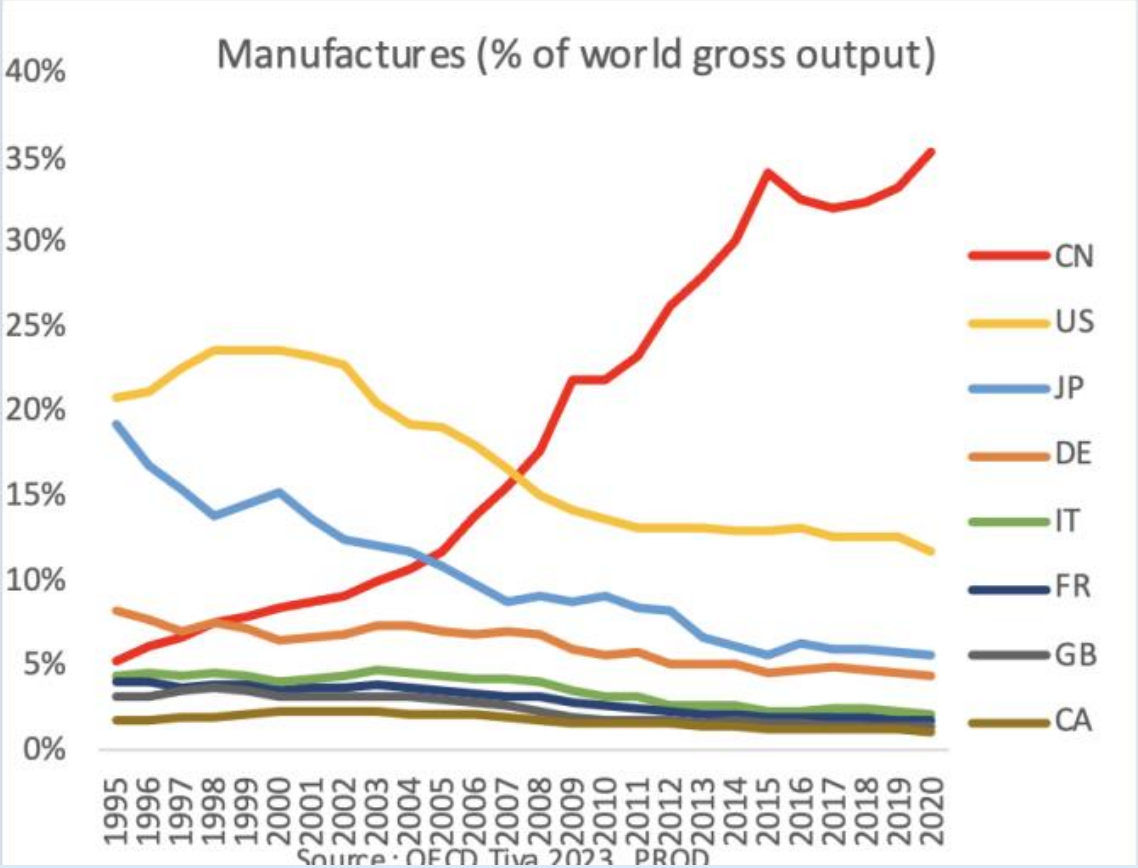
Town, *city*	Product	Town's claimed contribution
Ouhai, *Wenzhou*	Eyewear	90% of global-brand products (2016)
Jinjiang, *Quanzhou*	Zips	10% of global production (2011)
Chenghai, *Shantou*	Toys	30% of global production (2014)
Shiling, *Guangzhou*	Bags, suitcases	70% of European and US mass market (2013)
Gurao, *Shantou*	Bras	12% of China's production (2014)
Tangxia, *Dongguan*	Golf products	40% of global production (2011)

*Places indicated may include multiple clusters

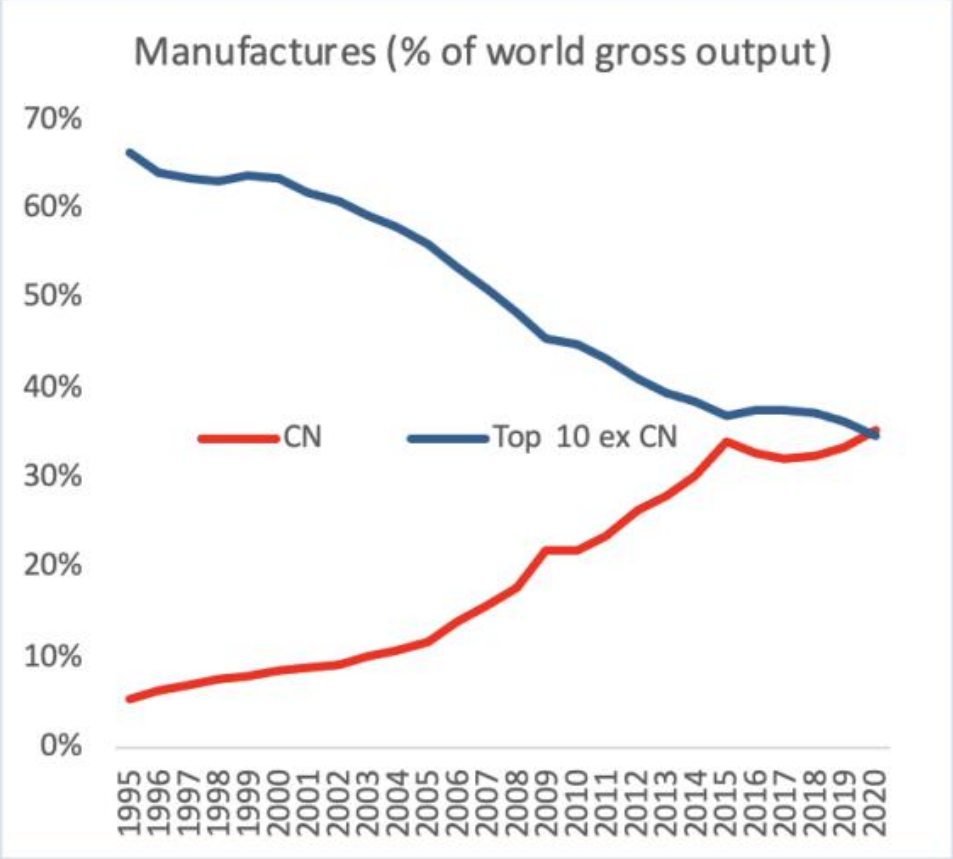
:conomist.com

World shares of gross production

G7 shares declined sharply (gross production)



China's share rose sharply (gross production)



Source: Baldwin (2024)

Table 4.1. Types of trade

Type of trade	Phrase	Meaning	
Inter-industry	Either/or	*Either* imports *or* exports in a given sector of the economy	North-South
Horizontal intra-industry	Both/and/same	*Both* imports *and* exports in a given sector of the economy and at the *same* stage of processing	North-North
Vertical intra-industry (Outsourcing by firms)	Both/and/different	*Both* imports *and* exports in a given sector of the economy and at *different* stages of processing	North-South

Why this happens?

Why is outsourcing concentrated in few developing countries?

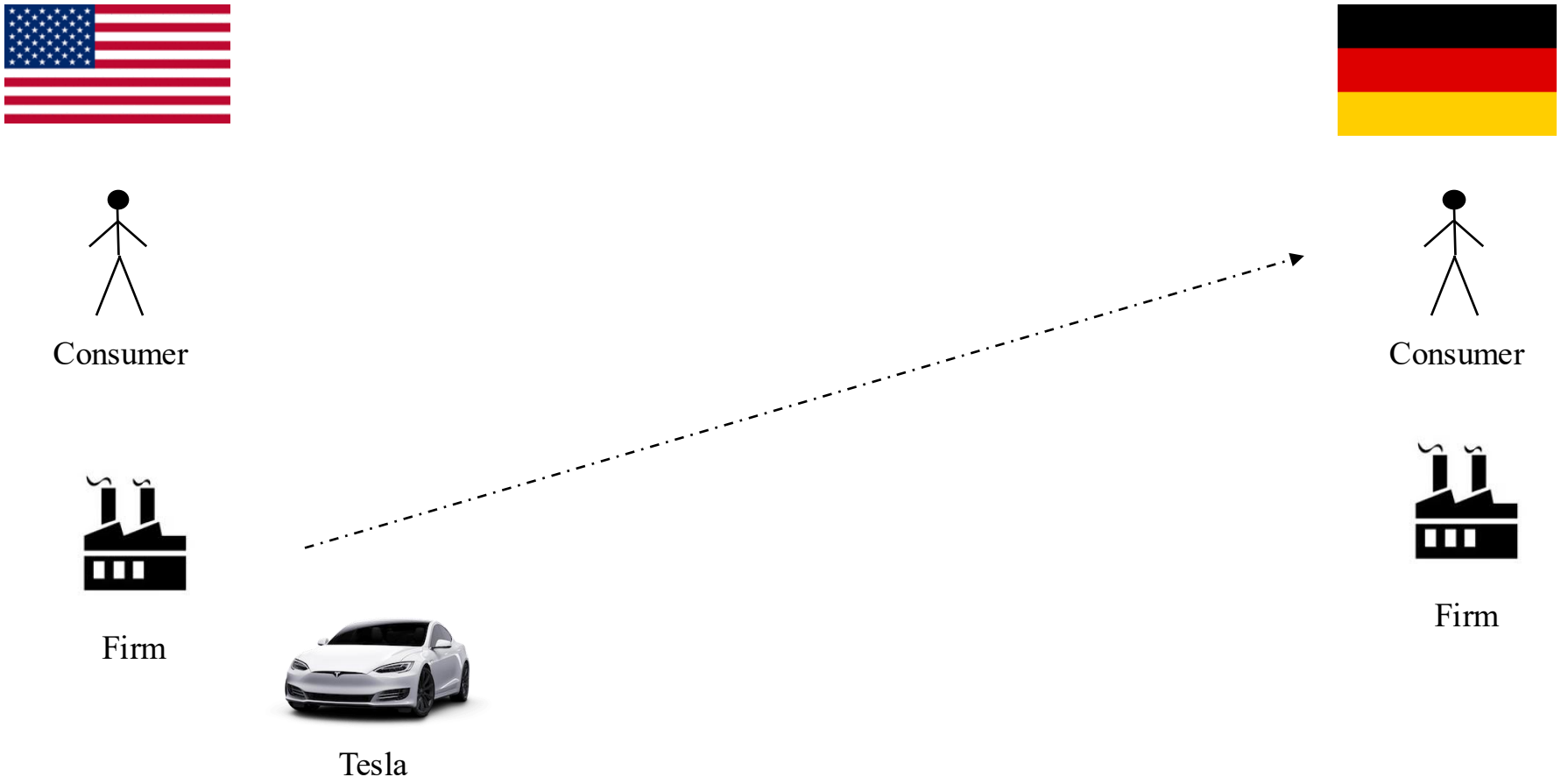
Lecture plan:

- New Trade Theory explaining North-North trade.
- New Trade Theory explaining concentration of outsourcing.
- Policy implications of New Trade Theory for “catch-up”.

Lecture plan:

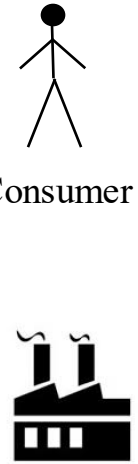
- **New Trade Theory explaining North-North trade.**
- New Trade Theory explaining concentration of outsourcing.
- Policy implications of New Trade Theory for “catch-up”.

Why the US beats Germany in electric car production?



Why the US beats Germany in electric car production?

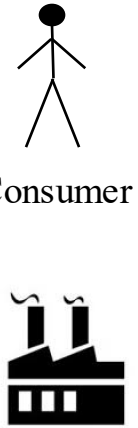
The US has been producing such cars at a bigger scale and for much longer (first mover advantage)



Consumer

Firm

Tesla



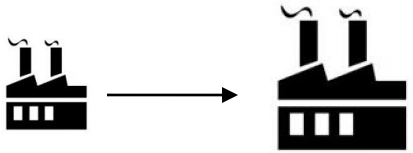
Consumer

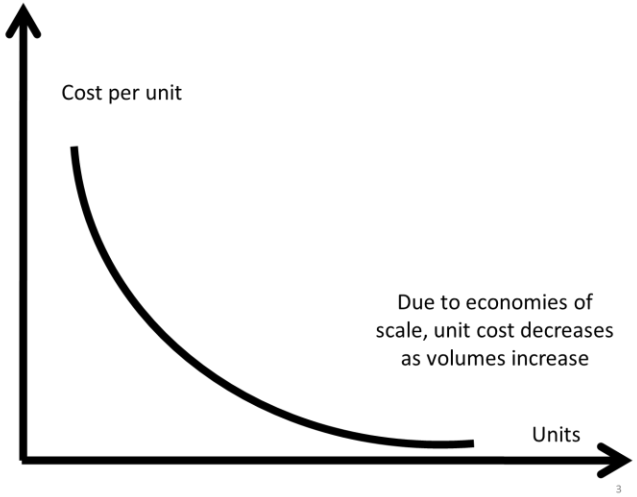
Firm

Difference in production costs → Price difference ← Differences in demand

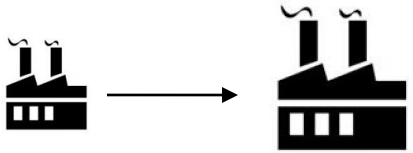
(Input costs, Technology, **Scale**)

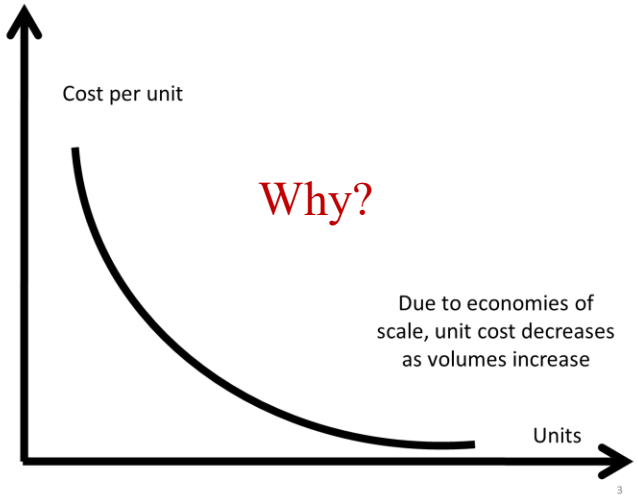
Internal economies of scale



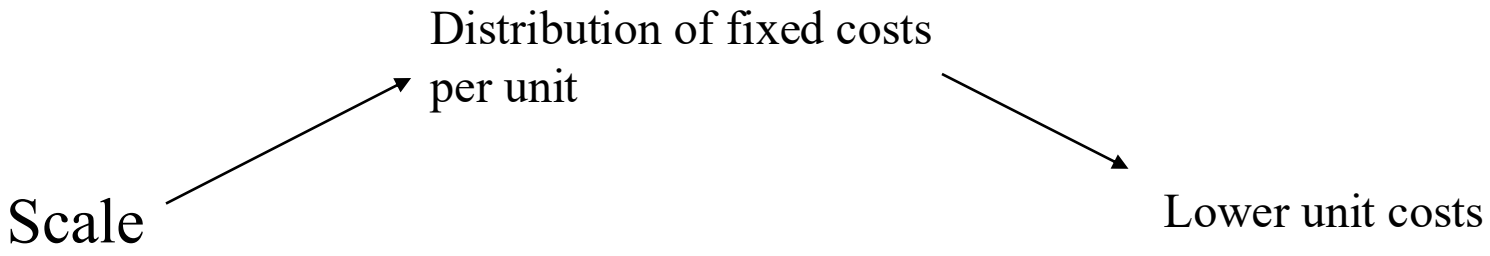


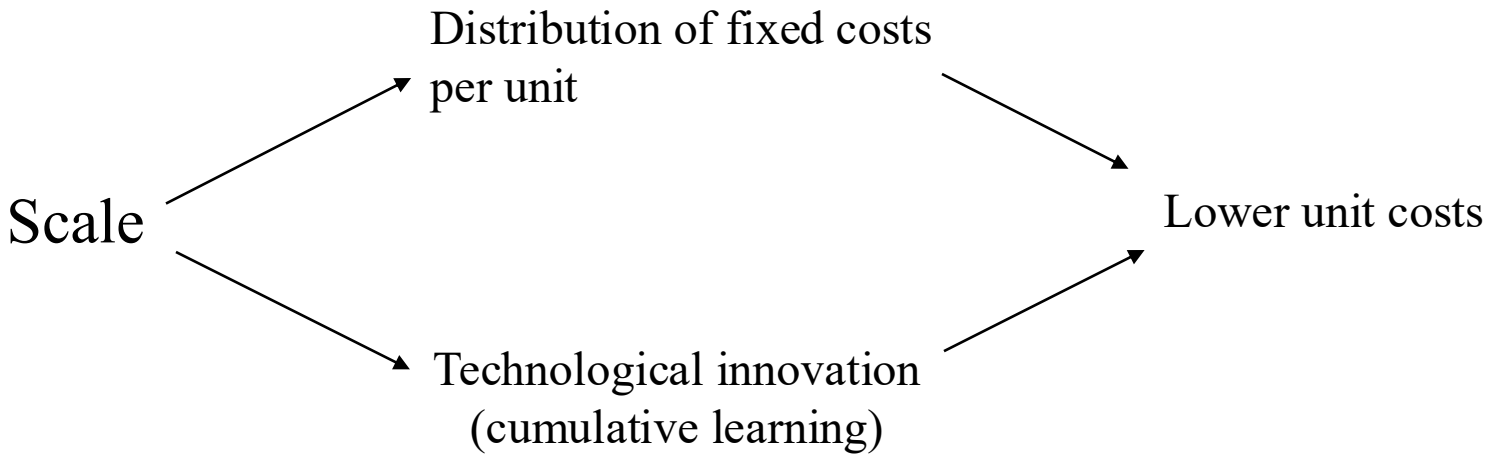
Internal economies of scale



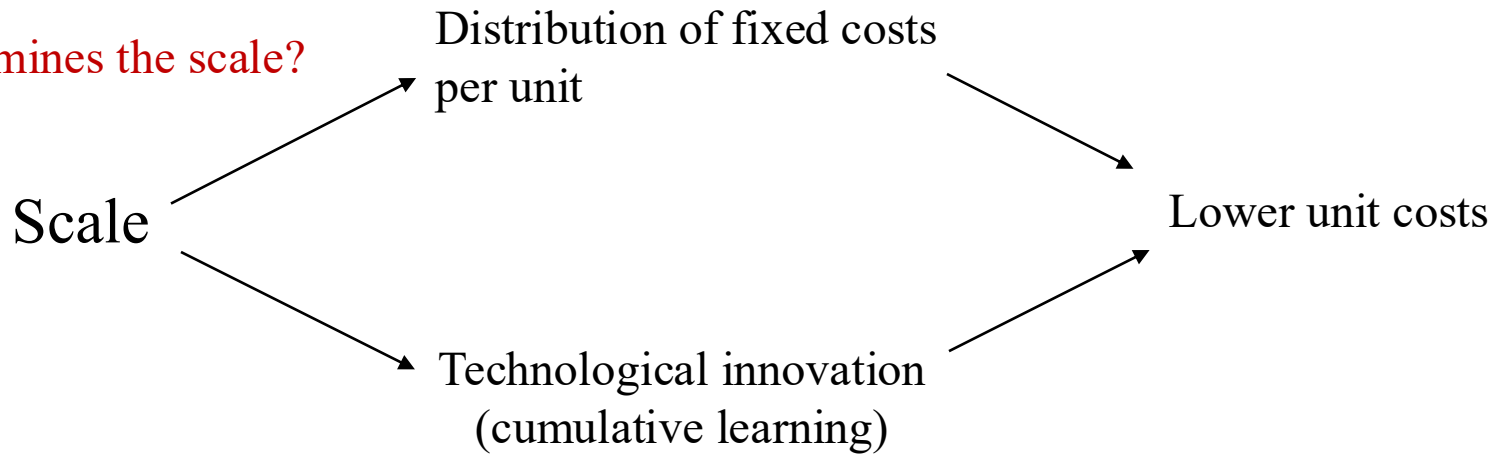


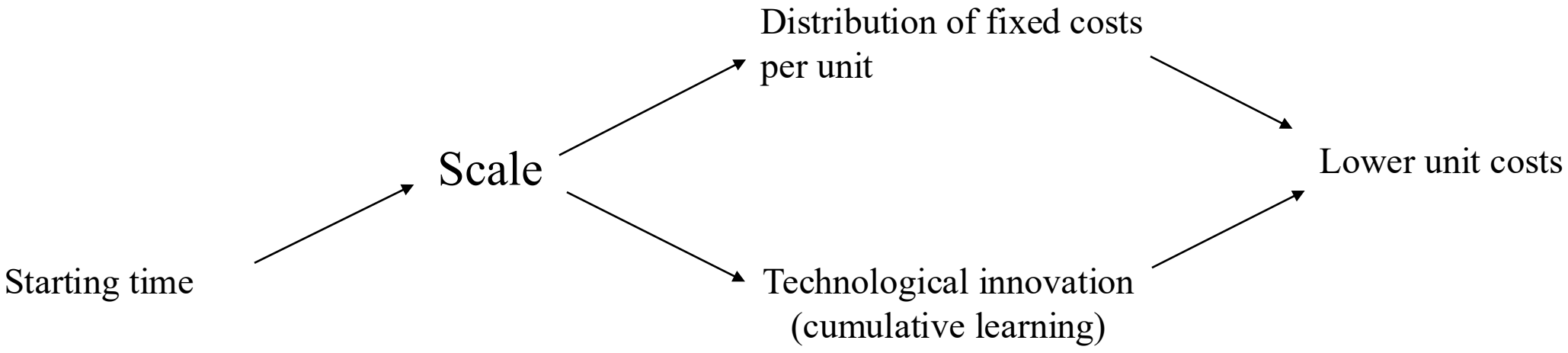
Scale

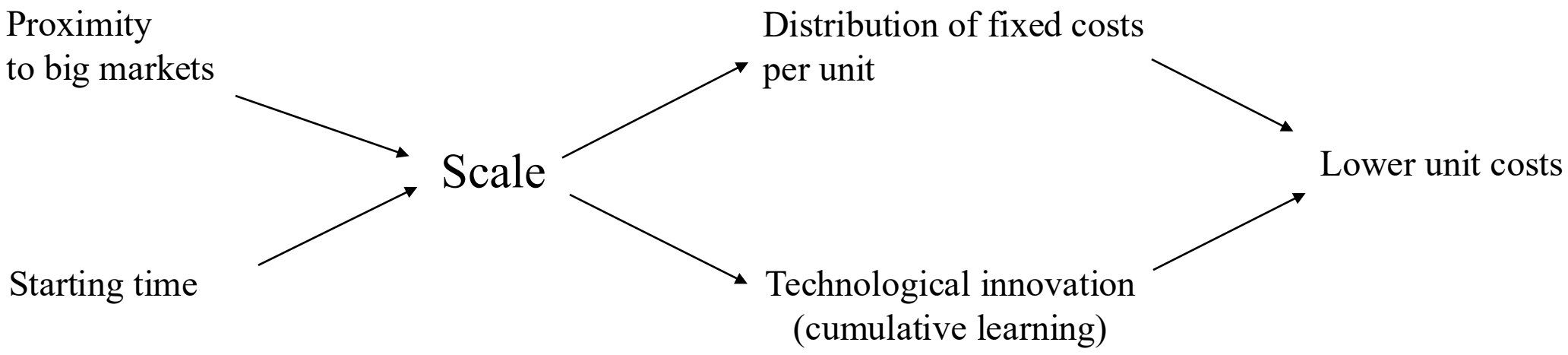

Distribution of fixed costs per unit
Scale
Lower unit costs


Distribution of fixed costs per unit
Scale
Lower unit costs
Technological innovation (cumulative learning)

What determines the scale?




Distribution of fixed costs per unit
Scale
Lower unit costs
Starting time
Technological innovation (cumulative learning)


Proximity
to big markets
Starting time
Scale
Distribution of fixed costs
per unit
Technological innovation
(cumulative learning)
Lower unit costs

Why does the US beat Germany in electric car production?

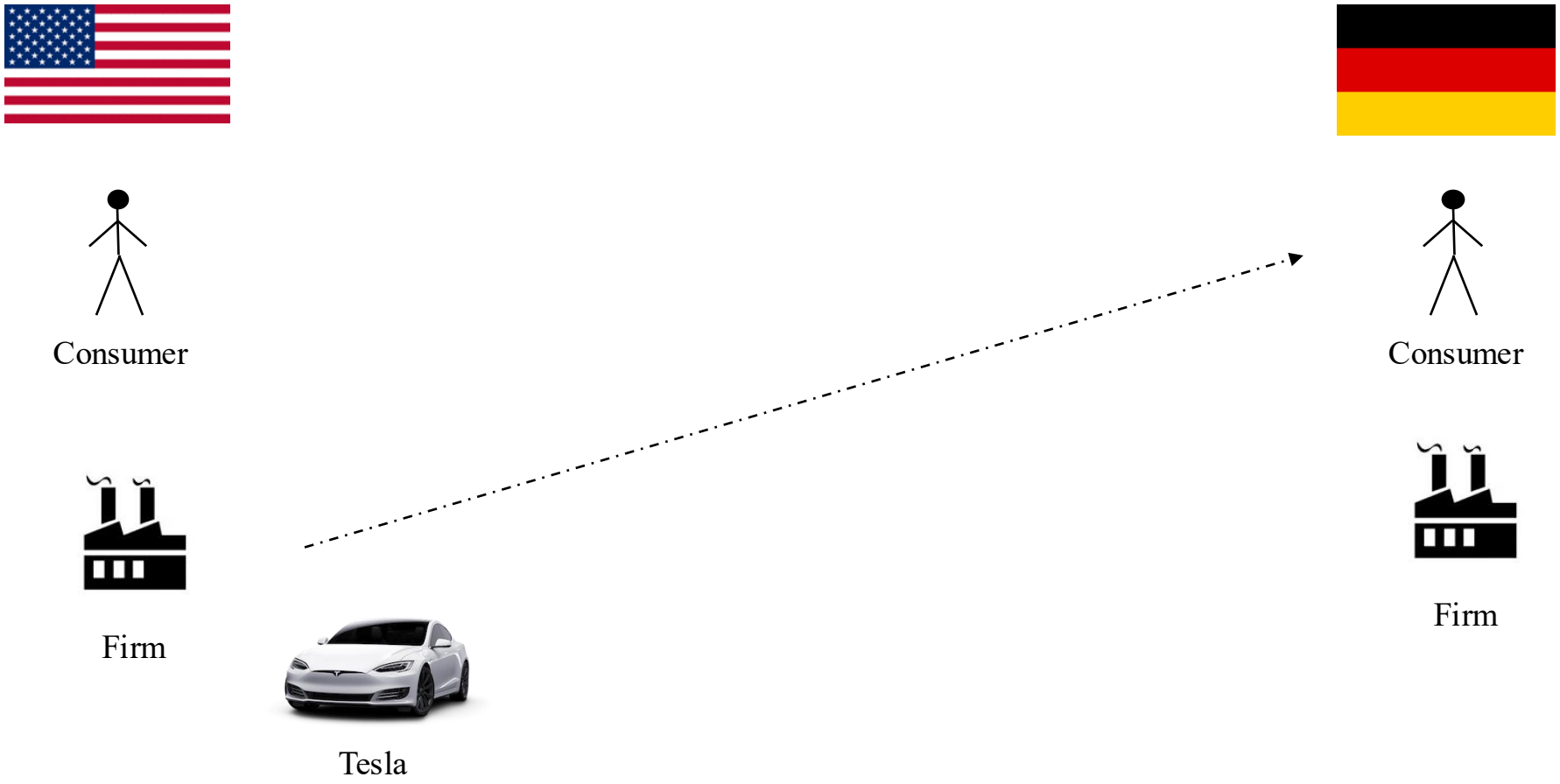


TABLE 6-2 Hypothetical Example of Gains from Market Integration

	Home Market, Before Trade	Foreign Market, Before Trade	Integrated Market, After Trade
Total sales of autos	900,000	1,600,000	2,500,000
Number of firms	6	8	10
Sales per firm	150,000	200,000	250,000
Average cost	$10,000	$8,750	$8,000
Price	$10,000	$8,750	$8,000

Source: P130 Krugman & Obstfled. 8th edition

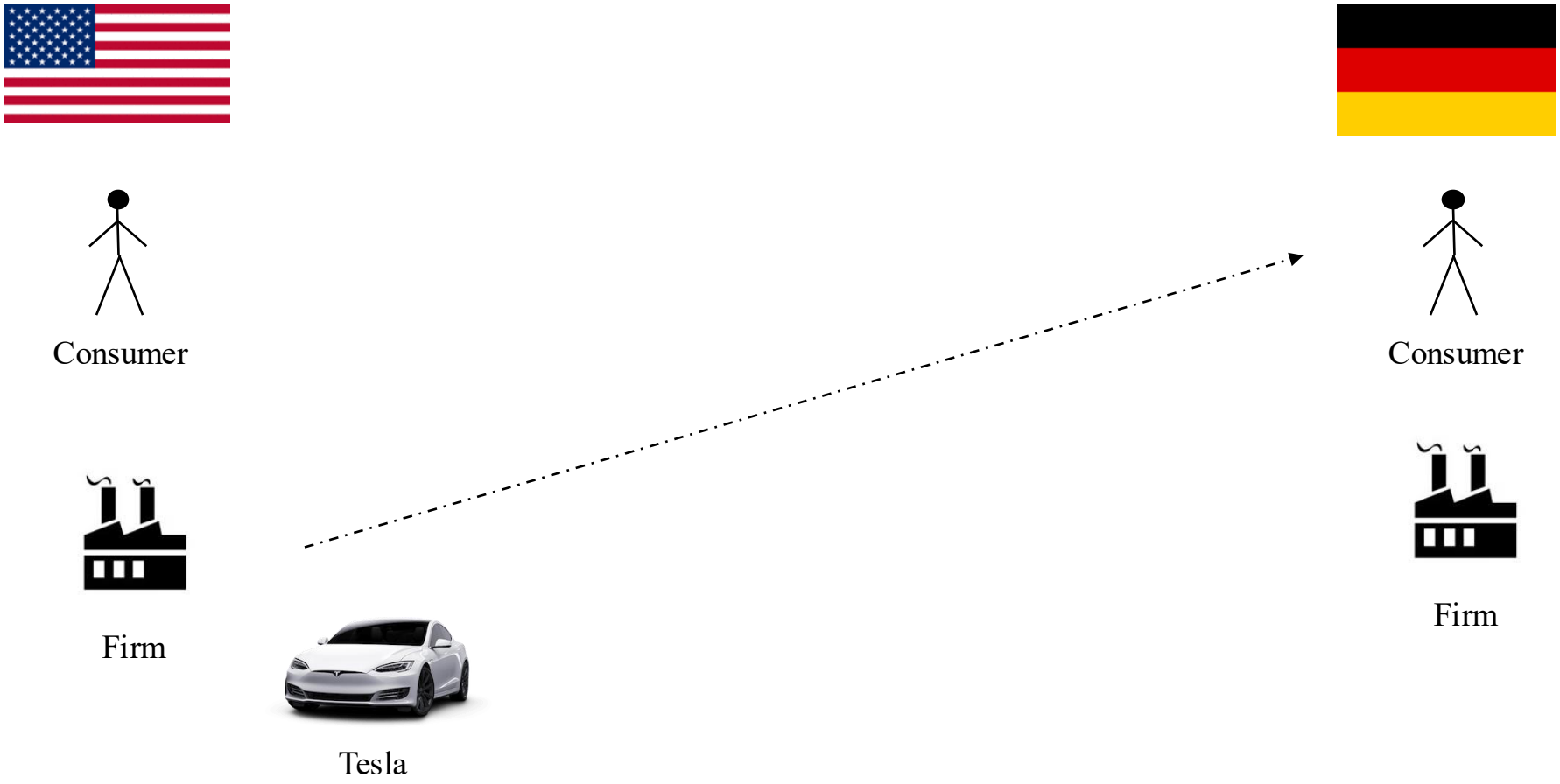
Electric cars	German market	US market
Total sales of autos	900,000	1,600,000
Number of firms	6	8
Sales per firm	150,000	200,000
Average cost	$10,000	$8,750
Price	$10,000	$8,750

Electric cars	German market	US market	Combined market
Total sales of autos	900,000	1,600,000	2,500,000
Number of firms	6	8	10
Sales per firm	150,000	200,000	250,000
Average cost	$10,000	$8,750	$8,000
Price	$10,000	$8,750	$8,000

Electric cars	German market	US market	Combined market
Total sales of autos	900,000	1,600,000	2,500,000
Number of firms	6	8	10
Sales per firm	150,000	200,000	250,000
Average cost	$10,000	$8,750	$8,000
Price	$10,000	$8,750	$8,000

Located in the US
(due to bigger initial scale)

But cars are not homogenous: differentiated products



But cars are not homogenous: differentiated products

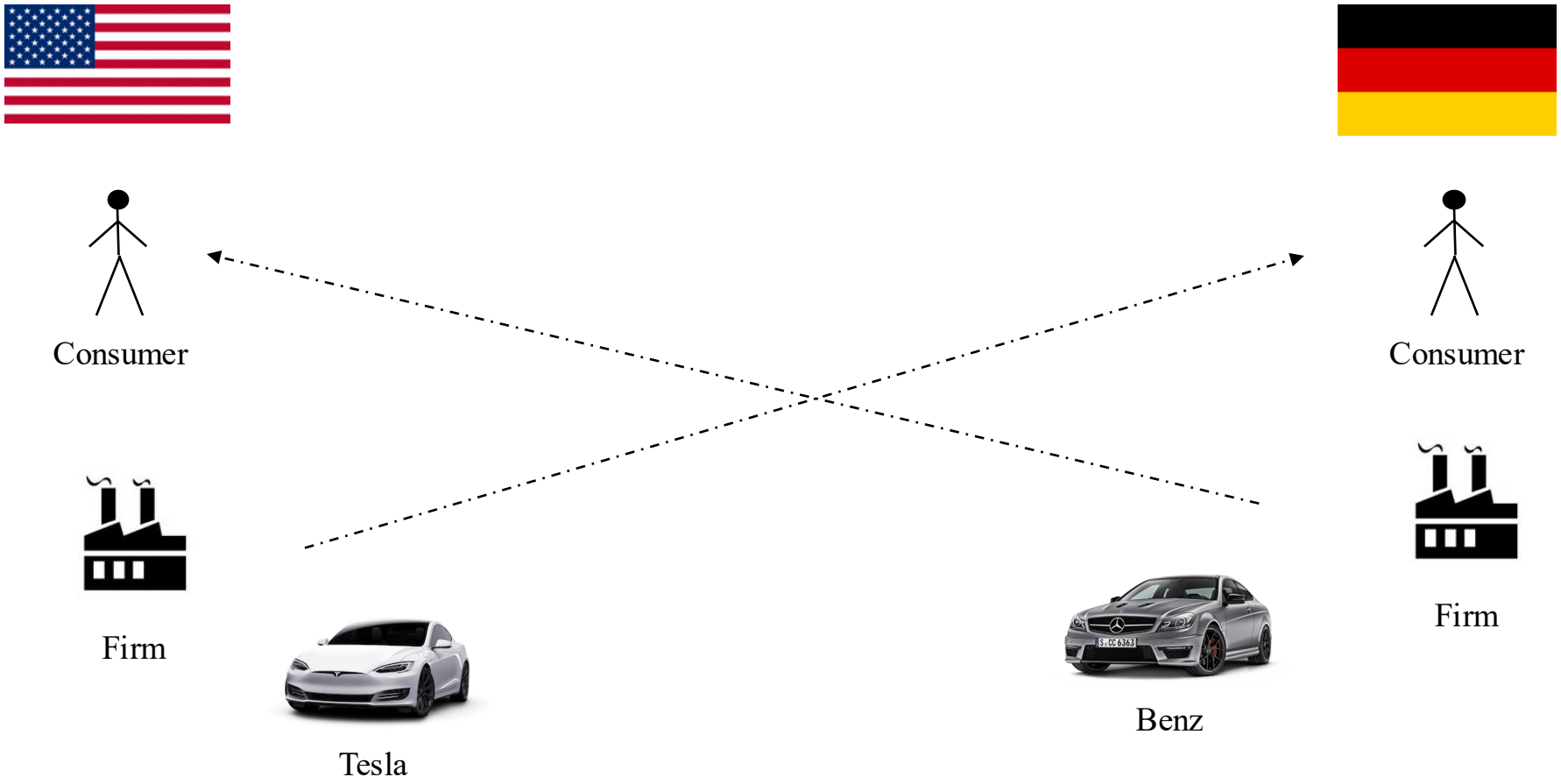


Table 4.1. Types of trade

Type of trade	Phrase	Meaning	
Inter-industry	Either/or	*Either* imports *or* exports in a given sector of the economy	North-South
Horizontal intra-industry	Both/and/same	*Both* imports *and* exports in a given sector of the economy and at the *same* stage of processing	North-North
Vertical intra-industry (Outsourcing by firms)	Both/and/different	*Both* imports *and* exports in a given sector of the economy and at *different* stages of processing	North-South

New Trade Theory

North-North trade: Rich countries gain scale and experience in one variety of a product and trade with each other, instead of producing all the varieties of that product.

Paul Krugman
(Nobel prize 2008)

Specialisation and trade allows countries to gain scale advantages in production of one variety of a product. This lowers the cost of production for all products and increases welfare at the global level.

Assumptions of the New Trade Theory:

Assumptions of the New Trade Theory:

Relative difference in production costs → Price difference

(Input costs, **Scale**, Technology)

Assumptions of the New Trade Theory:

Relative difference in production costs → Price difference

(Input costs, **Scale**, Technology)

- **Internal economies of scale (Imperfect competition).**

Assumptions of the New Trade Theory:

Relative difference in production costs → Price difference

(Input costs, **Scale**, Technology)

- Internal economies of scale (Imperfect competition).
- **Proximity to large markets matter (it helps you build scale).**

Assumptions of the New Trade Theory:

Relative difference in production costs → Price difference

(Input costs, **Scale**, Technology)

- Internal economies of scale (Imperfect competition).
- Proximity to large markets matter (it helps you build scale).
- **Technology improves with cumulative production (learning by doing), you can't simply import it.**

Assumptions of the New Trade Theory:

Relative difference in production costs → Price difference ← **Differences in demand**

(Input costs, **Scale**, Technology) **(product variety)**

- Internal economies of scale (Imperfect competition).
- Proximity to large markets matter (it helps you build scale).
- Technology improves with cumulative production (learning by doing), you can't simply import it.
- **Products are not homogenous; demand for product variety.**

Assumptions of the New Trade Theory:

Relative difference in production costs → Price difference ← Differences in demand

(Input costs, **Scale**, Technology)

(product variety)

- Internal economies of scale (Imperfect competition).
- Proximity to large markets matter (it helps you build scale).
- Technology improves with cumulative production (learning by doing), you can't simply import it.
- Products are not homogenous; demand for product variety.

Table 4.1. Types of trade

Type of trade	Phrase	Meaning	
Inter-industry	Either/or	*Either* imports *or* exports in a given sector of the economy	North-South
New Trade Theory Horizontal intra-industry	Both/and/same	*Both* imports *and* exports in a given sector of the economy and at the *same* stage of processing	**North-North** **South-South**
Vertical intra-industry (Outsourcing by firms)	Both/and/different	*Both* imports *and* exports in a given sector of the economy and at *different* stages of processing	North-South

Table 4.1. Types of trade

	Type of trade	Phrase	Meaning	
	Inter-industry	Either/or	*Either* imports *or* exports in a given sector of the economy	North-South
New Trade Theory	Horizontal intra-industry	Both/and/same	*Both* imports *and* exports in a given sector of the economy and at the *same* stage of processing	North-North **South-South**
	Vertical intra-industry (Outsourcing by firms)	Both/and/different	*Both* imports *and* exports in a given sector of the economy and at *different* stages of processing	North-South

Who are the winners and losers of North-North trade inside rich countries?

Table 4.1. Types of trade

Type of trade	Phrase	Meaning	
Inter-industry	Either/or	*Either* imports *or* exports in a given sector of the economy	North-South
New Trade Theory Horizontal intra-industry	Both/and/same	*Both* imports *and* exports in a given sector of the economy and at the *same* stage of processing	**North-North**
Vertical intra-industry (Outsourcing by firms)	Both/and/different	*Both* imports *and* exports in a given sector of the economy and at *different* stages of processing	North-South

Who are the winners and losers of North-North trade inside rich countries? Not clear. Adjustment takes place in the same sector and thus, it is less socially and politically painful.

Lecture plan:

- New Trade Theory explaining North-North trade.
- **New Trade Theory explaining concentration of outsourcing.**
- Policy implications of New Trade Theory for “catch-up”.

Why are Chinese industrial clusters so dominant in manufacturing?

Selected specialised clusters in southern China

Town, *city*	Product	Town's claimed contribution
Ouhai, *Wenzhou*	Eyewear	90% of global-brand products (2016)
Jinjiang, *Quanzhou*	Zips	10% of global production (2011)
Chenghai, *Shantou*	Toys	30% of global production (2014)
Shiling, *Guangzhou*	Bags, suitcases	70% of European and US mass market (2013)
Gurao, *Shantou*	Bras	12% of China's production (2014)
Tangxia, *Dongguan*	Golf products	40% of global production (2011)

Why are Chinese industrial clusters so dominant in manufacturing?

Selected specialised clusters in southern China

Town, *city*	Product	Town's claimed contribution
Ouhai, *Wenzhou*	Eyewear	90% of global-brand products (2016)
Jinjiang, *Quanzhou*	Zips	10% of global production (2011)
Chenghai, *Shantou*	Toys	30% of global production (2014)
Shiling, *Guangzhou*	Bags, suitcases	70% of European and US mass market (2013)
Gurao, *Shantou*	Bras	12% of China's production (2014)
Tangxia, *Dongguan*	Golf products	40% of global production (2011)

Low wages (favorable factor endowment) can't be the answer

Why are Chinese industrial clusters so dominant in manufacturing?

Selected specialised clusters in southern China

Town, *city*	Product	Town's claimed contribution
Ouhai, *Wenzhou*	Eyewear	90% of global-brand products (2016)
Jinjiang, *Quanzhou*	Zips	10% of global production (2011)
Chenghai, *Shantou*	Toys	30% of global production (2014)
Shiling, *Guangzhou*	Bags, suitcases	70% of European and US mass market (2013)
Gurao, *Shantou*	Bras	12% of China's production (2014)
Tangxia, *Dongguan*	Golf products	40% of global production (2011)

Low wages (favorable factor endowment) can't be the answer

Clusters create external economies of scale.

Canary Wharf

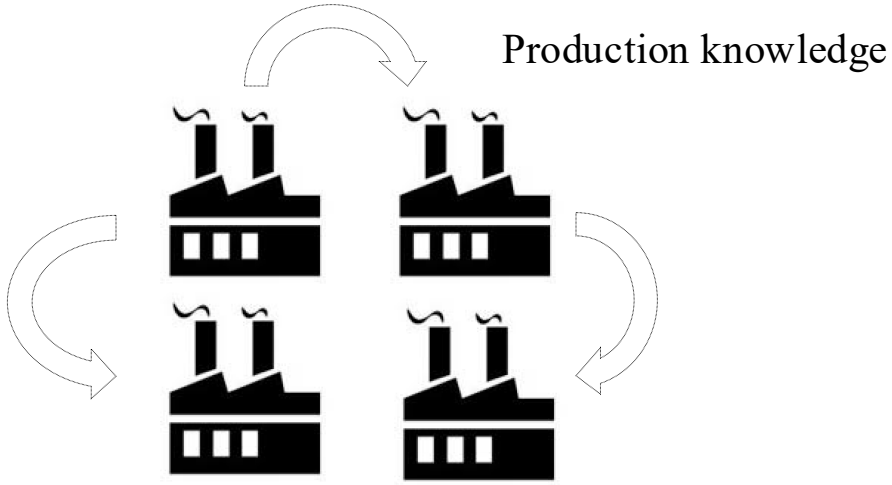
Shenzen

Bloomsbury Circle

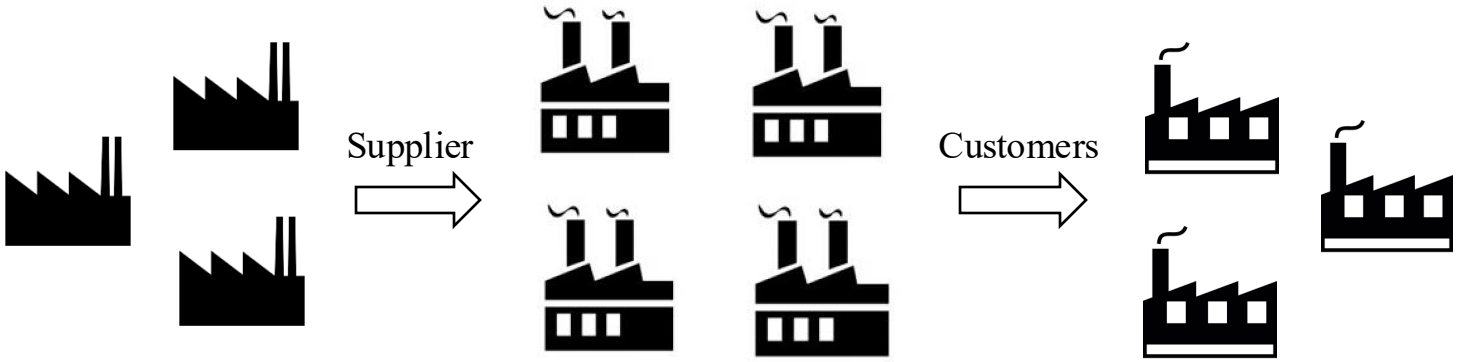
What are the benefit of firms clustering together? Why do firms locate next to each other in cities?

As competitors, suppliers and distributors get together the unit cost of production goes down (*external* economies of scale)

As competitors, suppliers and distributors get together the unit cost of production goes down (*external* economies of scale)



As competitors, suppliers and distributors get together the unit cost of production goes down (*external* economies of scale)



As competitors, suppliers and distributors get together the unit cost of production goes down (*external* economies of scale)

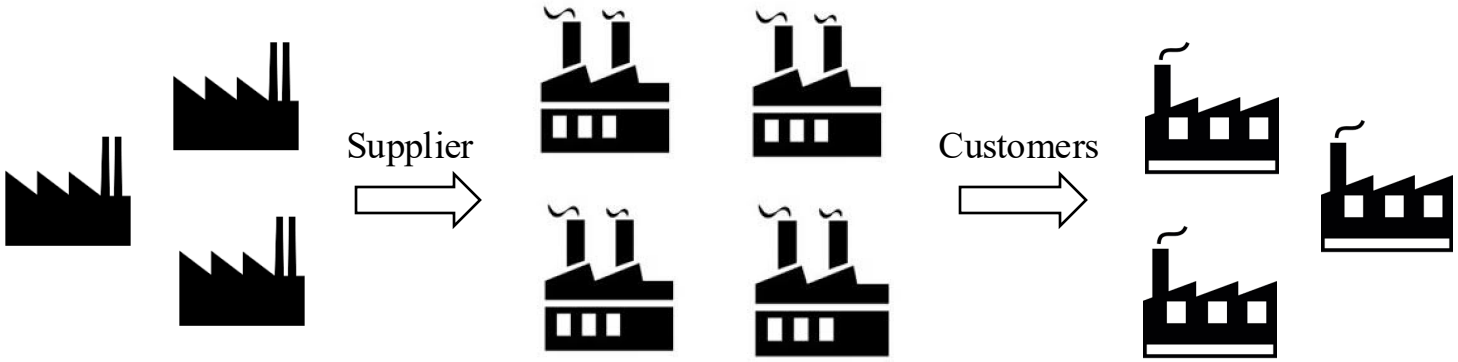


Table 4.1. Types of trade

Type of trade	Phrase	Meaning	
Inter-industry	Either/or	*Either* imports *or* exports in a given sector of the economy	North-South
Horizontal intra-industry	Both/and/same	*Both* imports *and* exports in a given sector of the economy and at the *same* stage of processing	North-North
Vertical intra-industry (Outsourcing by firms)	Both/and/different	*Both* imports *and* exports in a given sector of the economy and at *different* stages of processing	North-South

Why is outsourcing concentrated in few developing countries?

New Trade Theory: Because those regions/countries have developed vibrant industrial clusters while others have failed to do so.

Assumptions of the New Trade Theory:

Relative difference in production costs → Price difference ← Differences in demand

(Input costs, **Scale**, Technology)

(product variety)

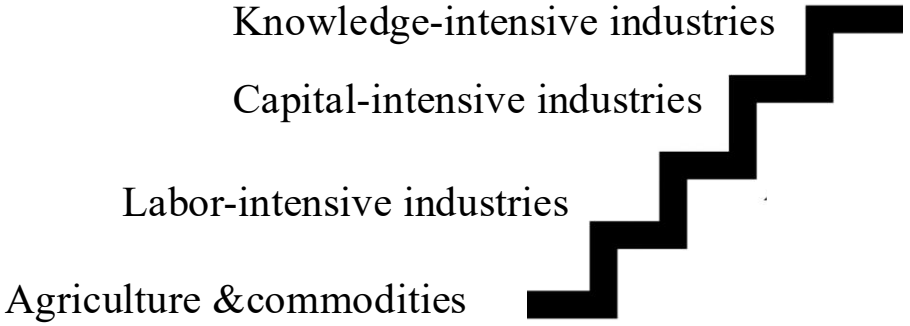
- **External economies of scale.**
- Proximity to large markets matter (it helps you build scale).
- Technology improves with cumulative production (learning by doing), you can't simply import it.
- Products are not homogenous; demand for product variety.

Krugman Rose New Trade Theory

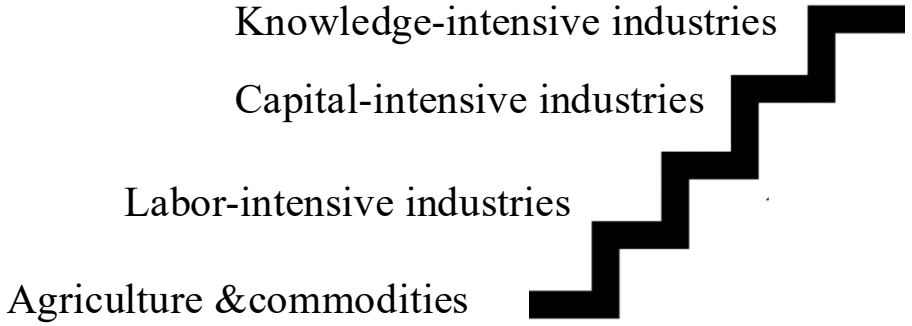
CLICK HERE

Lecture plan:

- New Trade Theory explaining North-North trade.
- New Trade Theory explaining concentration of outsourcing.
- **Policy implications of New Trade Theory for “catch-up”.**

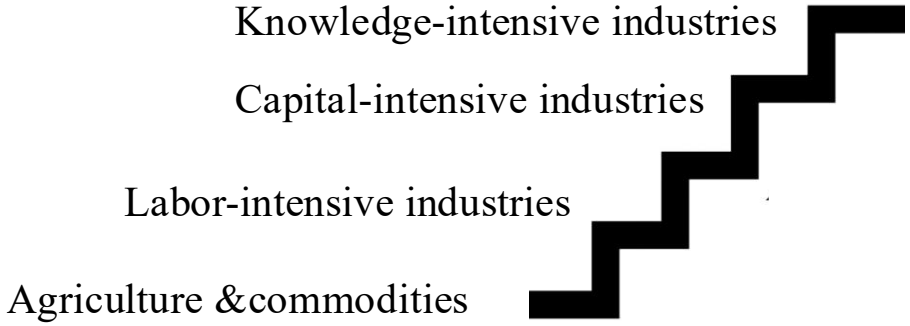

Knowledge-intensive industries
Capital-intensive industries
Labor-intensive industries
Agriculture &commodities

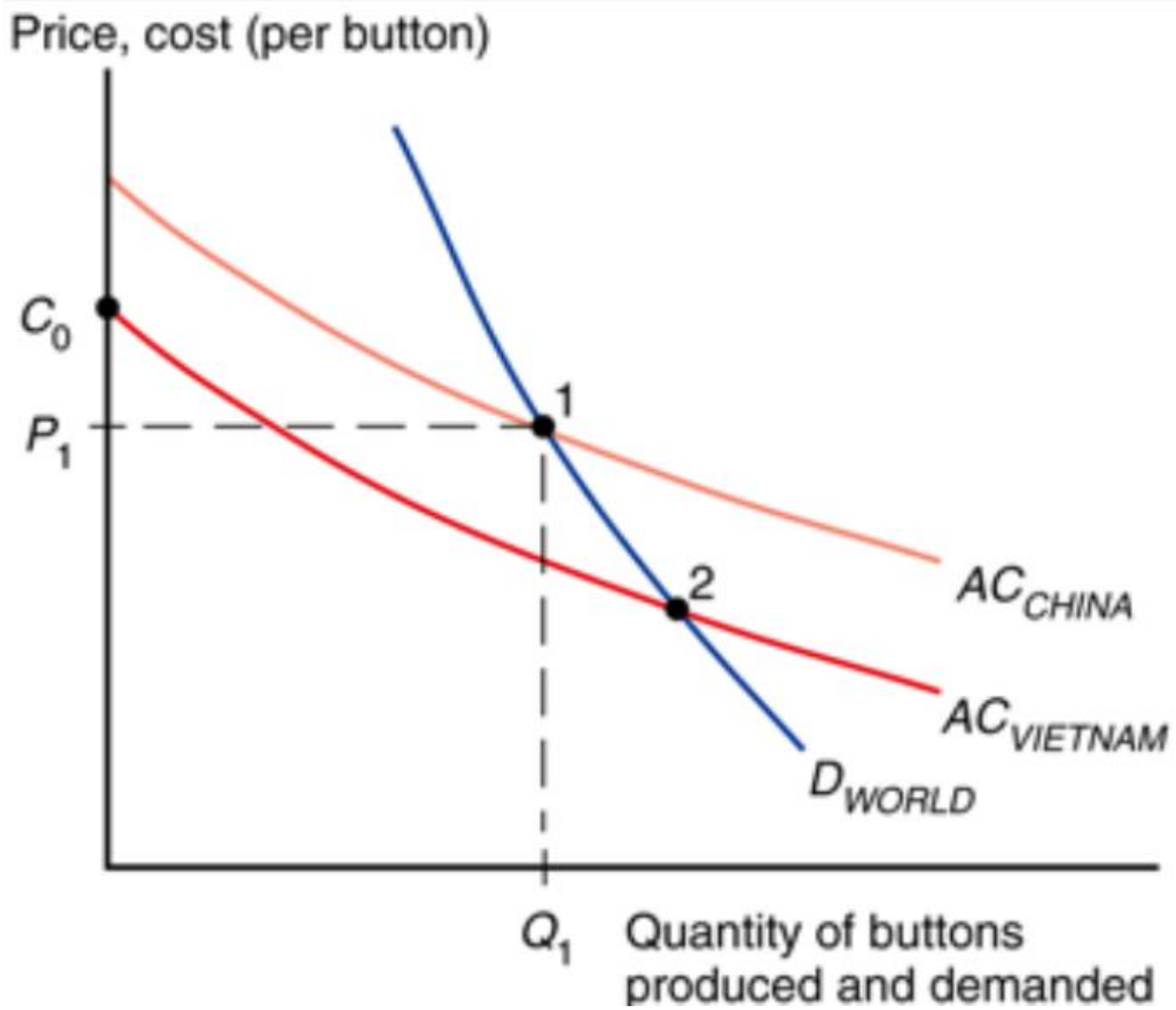
-Having the right factor endowments (potential comparative advantage) isn’t enough to compete in a sector.

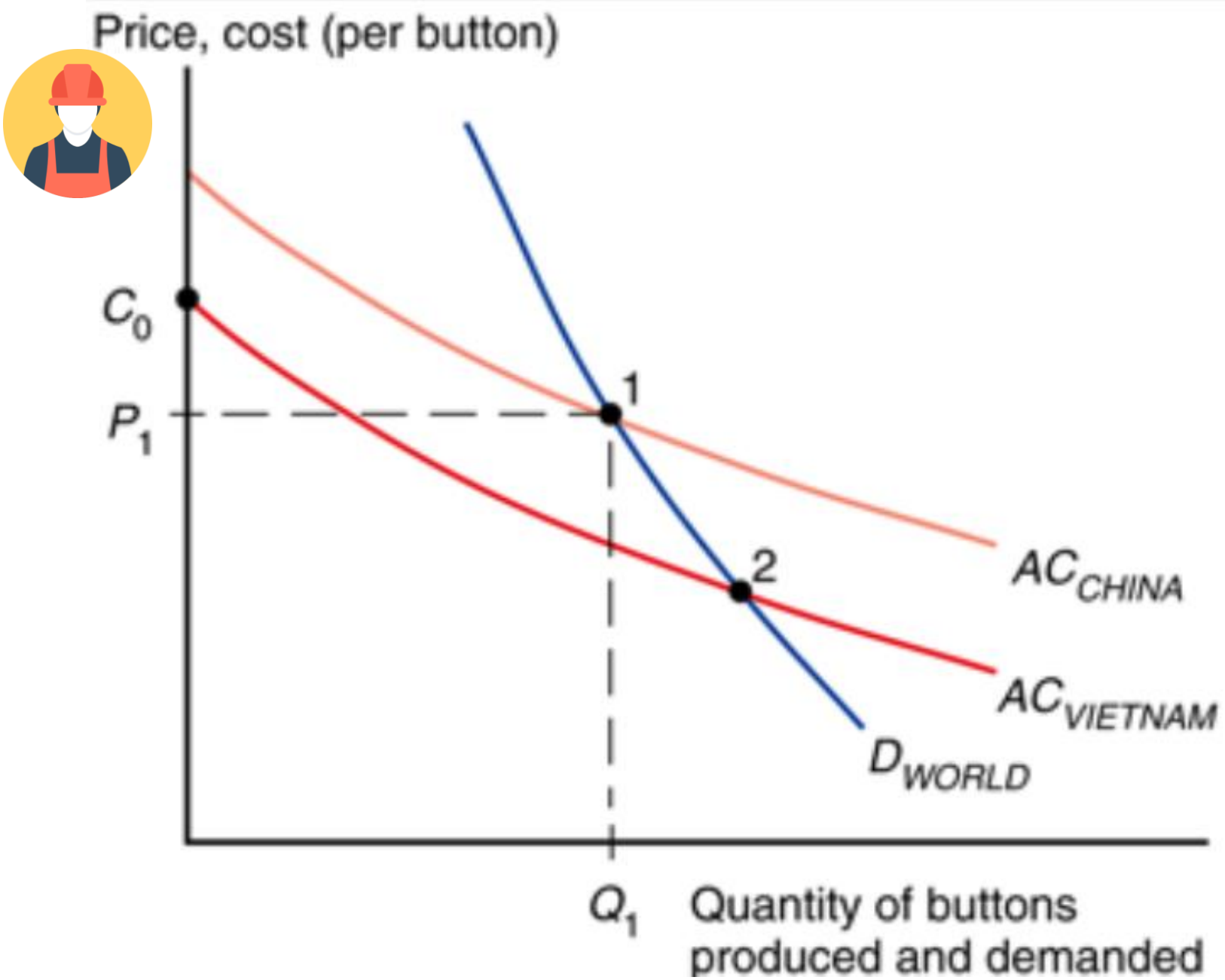


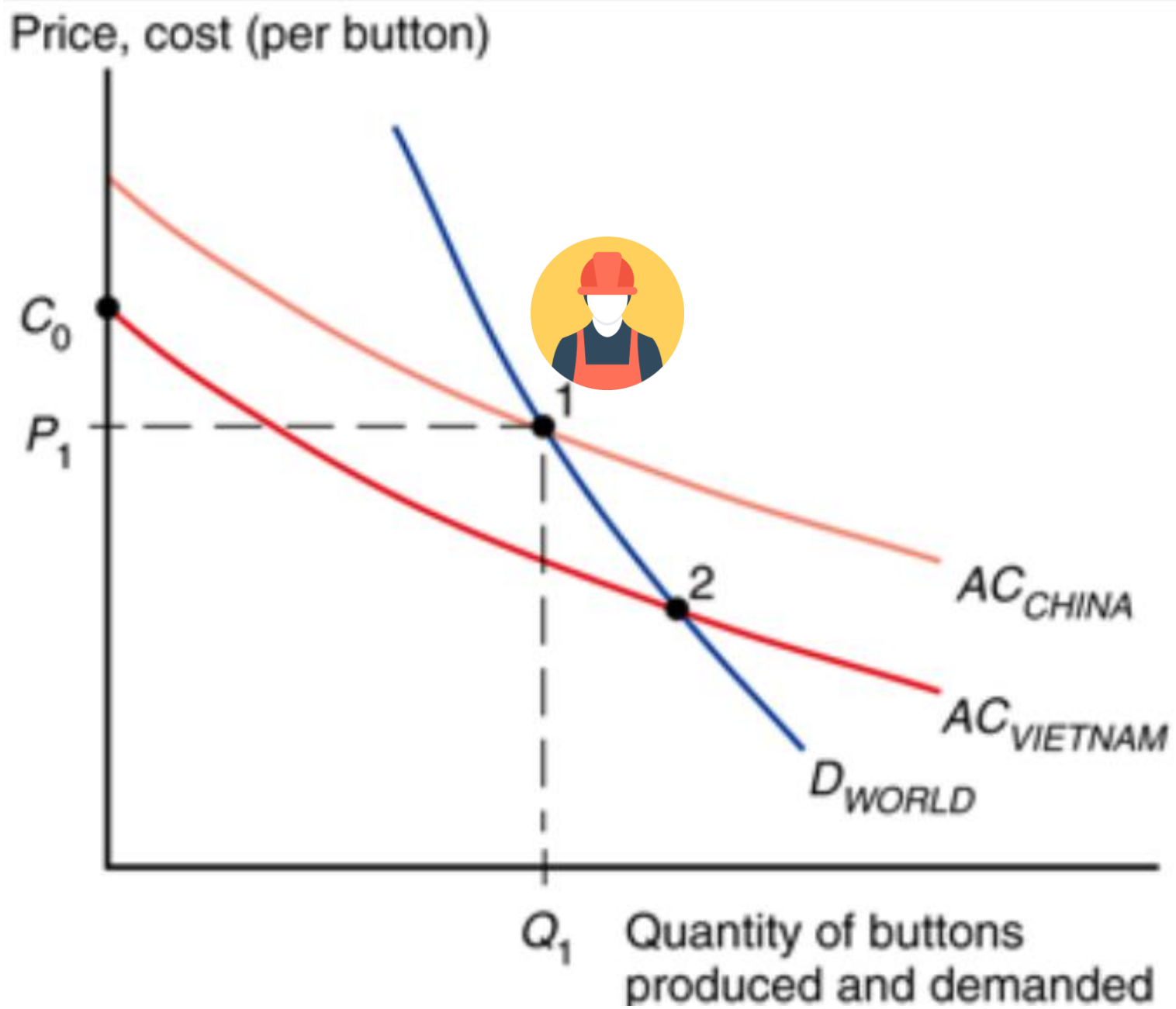
-Having the right factor endowments (potential comparative advantage) isn’t enough to compete in a sector.

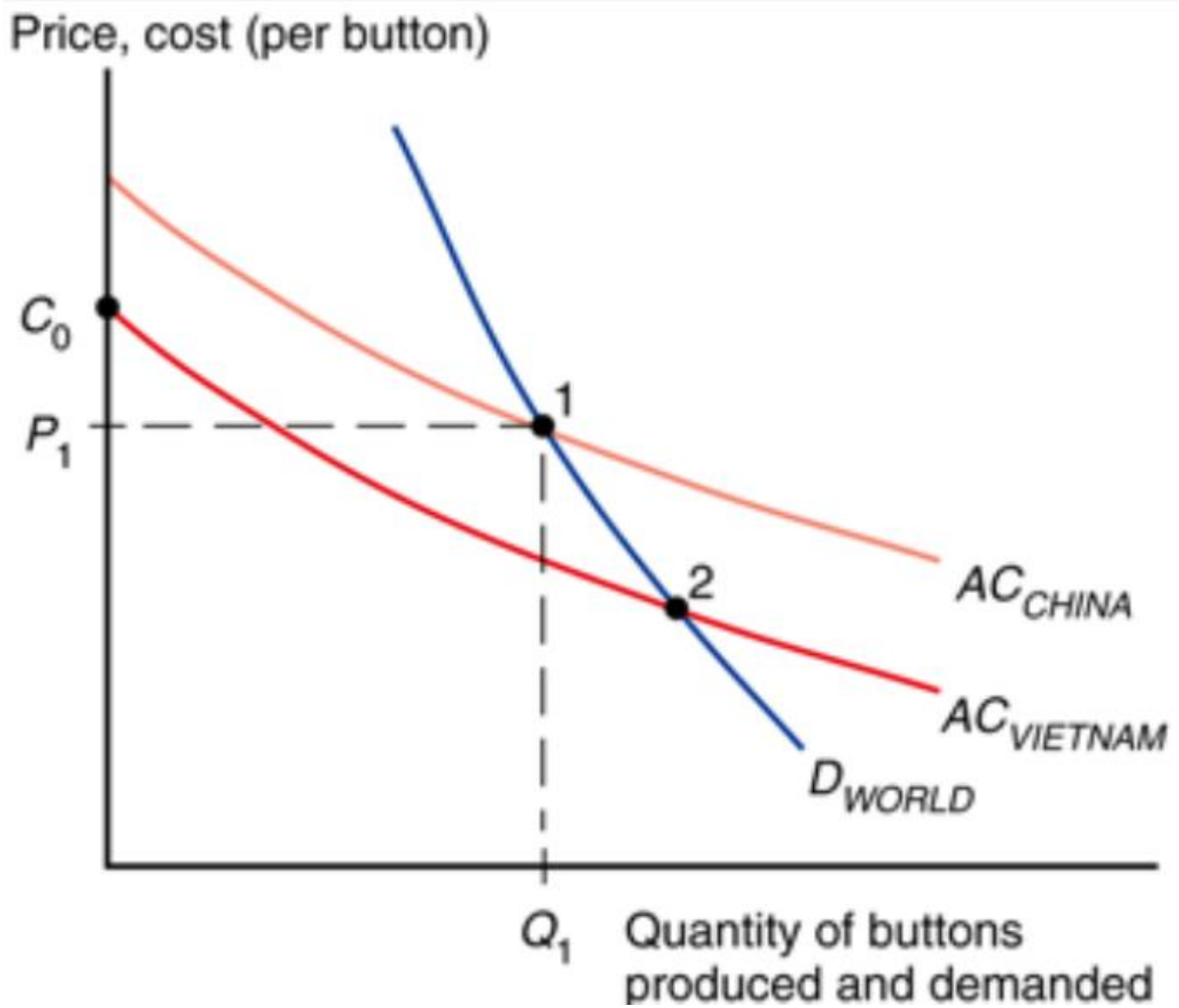
-Scale disadvantage and knowledge disadvantage for latecomers.




Price, cost (per button)
C_0
P_1
1
2
AC_{CHINA}
$AC_{VIETNAM}$
D_{WORLD}
Q_1
Quantity of buttons produced and demanded


Price, cost (per button)
C_0
P_1
1
2
AC_{CHINA}
$AC_{VIETNAM}$
D_{WORLD}
Q_1
Quantity of buttons produced and demanded


Price, cost (per button)
C_0
P_1
1
2
AC_{CHINA}
$AC_{VIETNAM}$
D_{WORLD}
Q_1
Quantity of buttons produced and demanded

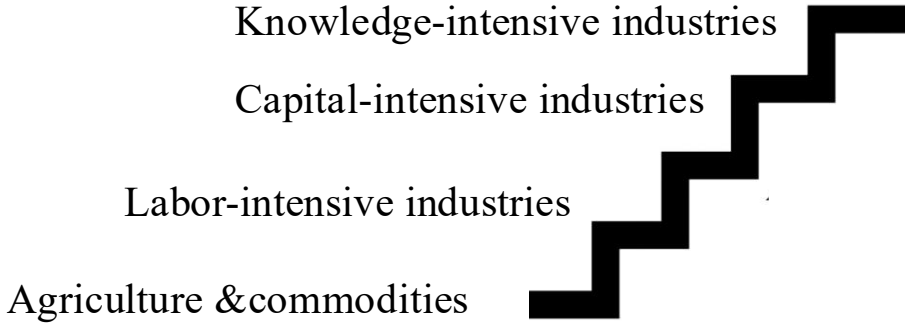


Vietnam can potentially produce buttons more cheaper than China (lower wages). But if Chinese firms/clusters are established earlier then they are able to sell buttons at p1, which is below the cost of production for the initial Vietnamese firms (Co). So latecomers are disadvantaged.

-Having the right factor endowments (potential comparative advantage) isn't enough to compete in a sector.

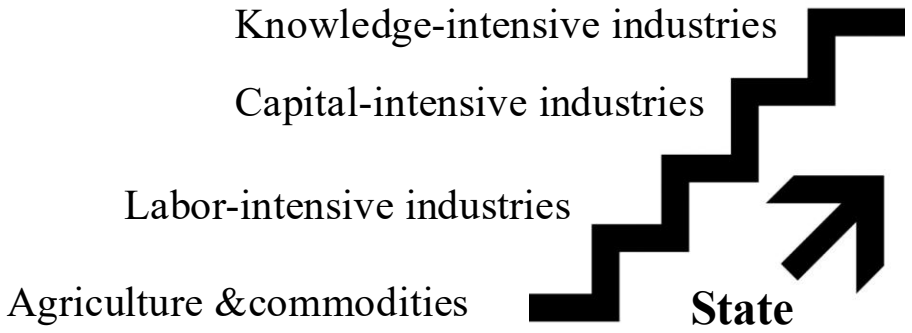
-Scale disadvantage and knowledge disadvantage for latecomers.

Solution?



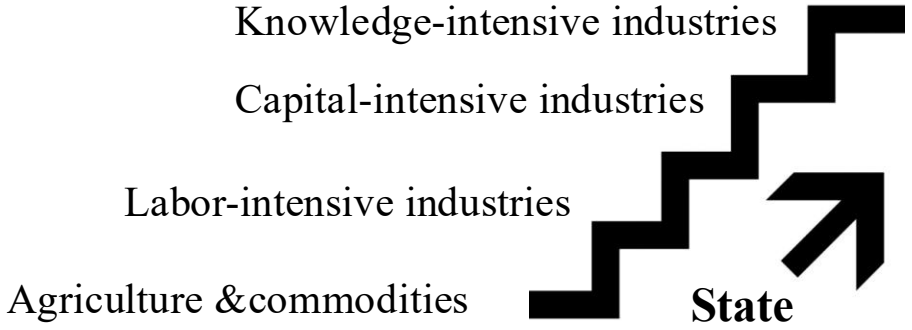
-Having the right factor endowments (potential comparative advantage) isn't enough to compete in a sector.

-Scale disadvantage and knowledge disadvantage for latecomers.



Solution? The state has to support infant firms in sectors in which the country has a comparative advantage.

Policy tools?



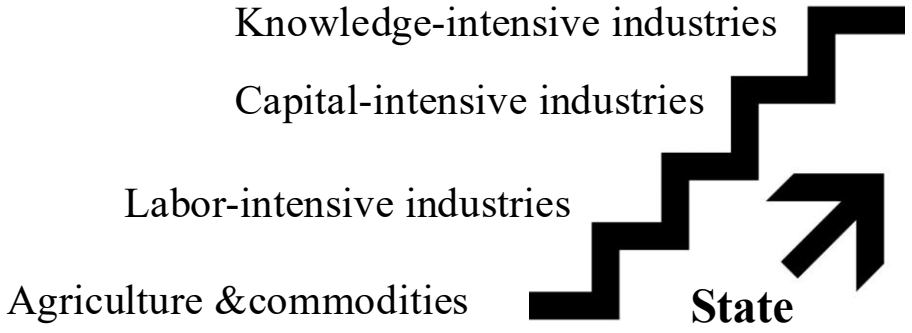
Policy tools:

Tariff protection

Credit subsidies

Tax breaks

Infrastructure(SEZ)



China's Huajian Group boost Ethiopian manufacturing sector

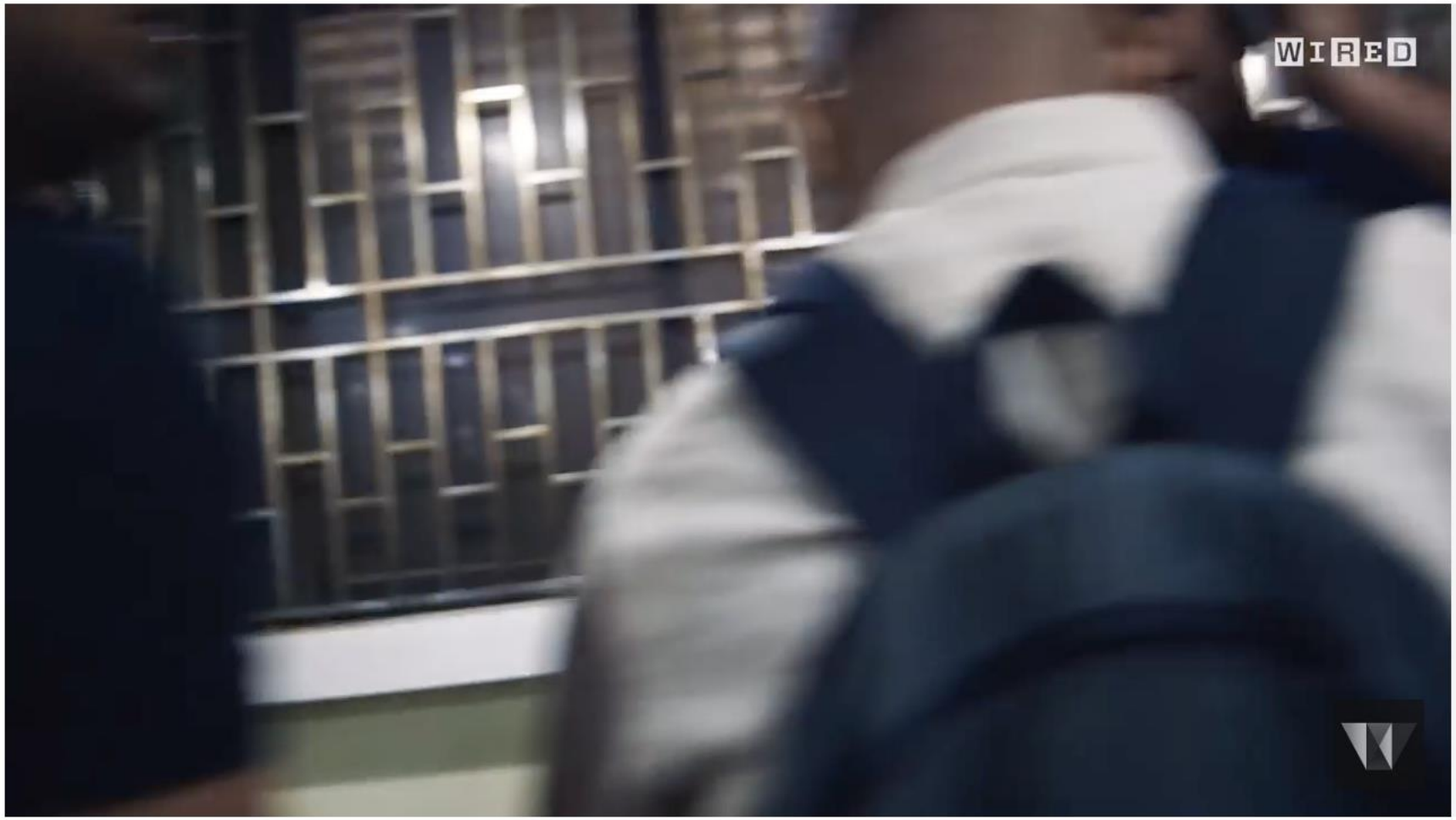
CGTN

Subscribe 112,334

9,897 views

Download Add to Share More 34 2

CLICK HERE



Shenzhen: The Silicon Valley of Hardware (Full Documentary) | Future Cities | WIRED

CLICK HERE

Sorry, a bit of self-promotion

Inside UCL Economics videos

Watch on our Instagram

@UCLECONOMICS

Summary

Summary

- New Trade Theory explains North-North trade. Similar countries trade produce a different variety of a product because through specialization they accumulate scale and production knowledge.

Summary

- New Trade Theory explains North-North trade. Similar countries trade produce a different variety of a product because through specialization they accumulate scale and production knowledge.

- North-North trade entails less painful labor market adjustments.

Summary

- New Trade Theory explains North-North trade. Similar countries trade produce a different variety of a product because through specialization they accumulate scale and production knowledge.

- North-North trade entails less painful labor market adjustments.

- New Trade Theory explain why the majority of North-South outsourcing happens in few industrial clusters in developing countries.

Summary

- New Trade Theory explains North-North trade. Similar countries trade produce a different variety of a product because through specialization they accumulate scale and production knowledge.

- North-North trade entails less painful labor market adjustments.

- New Trade Theory explain why the majority of North-South outsourcing happens in few industrial clusters in developing countries.

- New Trade Theory's policy implication for "catch-up": infant industry support.

Summary

- New Trade Theory explains North-North trade. Similar countries trade produce a different variety of a product because through specialization they accumulate scale and production knowledge.

- North-North trade entails less painful labor market adjustments.

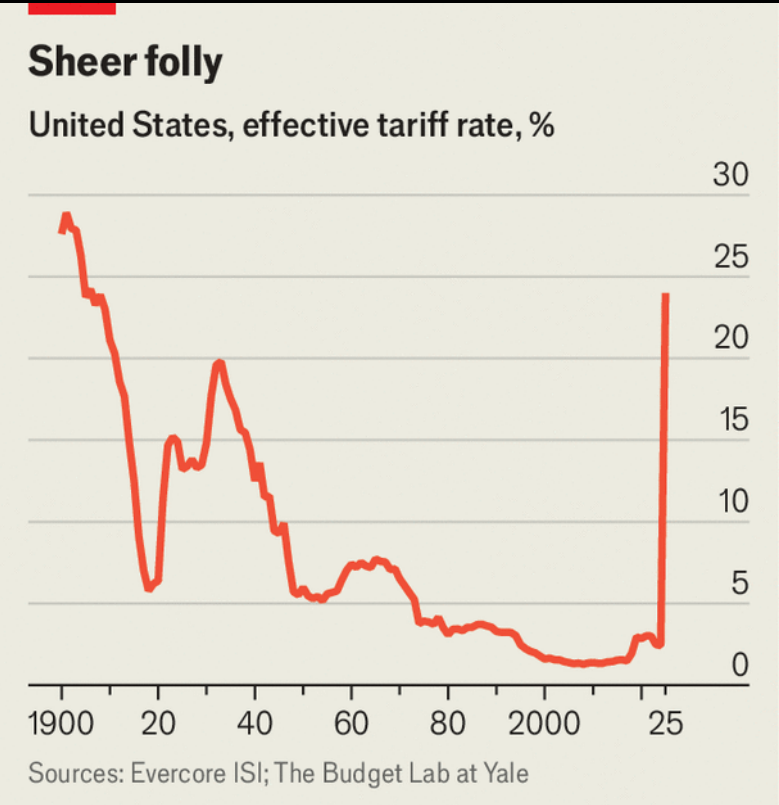
- New Trade Theory explain why the majority of North-South outsourcing happens in few industrial clusters in developing countries.

- New Trade Theory's policy implication for "catch-up": infant industry support. Hence, the need for flexible globalisation that leaves room for national policymaking.

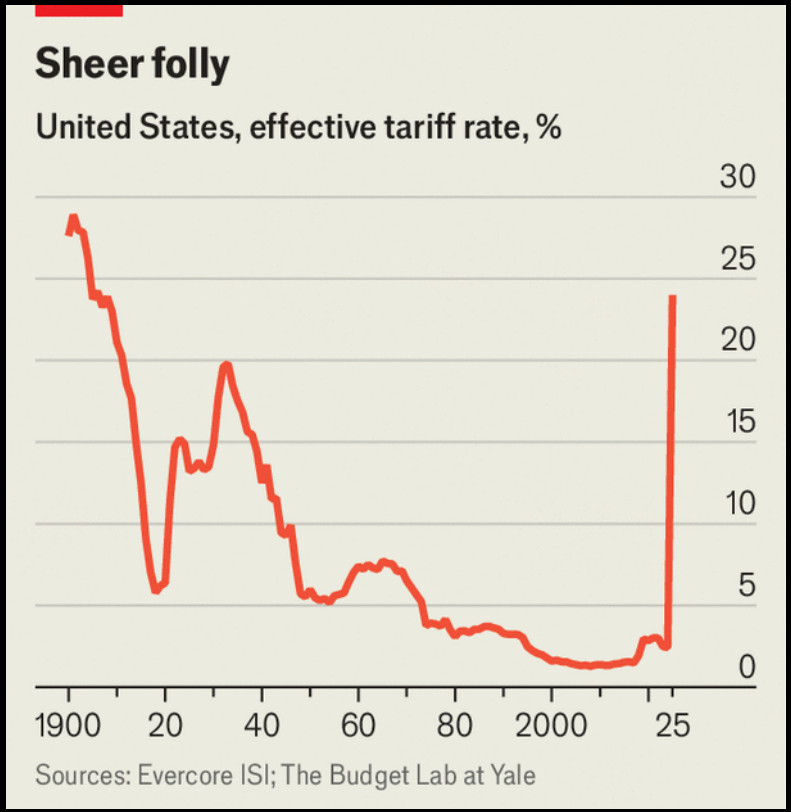
Table 4.1. Types of trade

	Type of trade	Phrase	Meaning
AA explains HO explains	Inter-industry	Either/or	*Either* imports *or* exports in a given sector of the economy
NTT explains	Horizontal intra-industry	Both/and/same	*Both* imports *and* exports in a given sector of the economy and at the *same* stage of processing
AA explains HO explains	Vertical intra-industry (Outsourcing by firms)	Both/and/different	*Both* imports *and* exports in a given sector of the economy and at *different* stages of processing
NTT explains	Why outsourcing is concentrated in few developing countries?		

Why are tariffs bad?

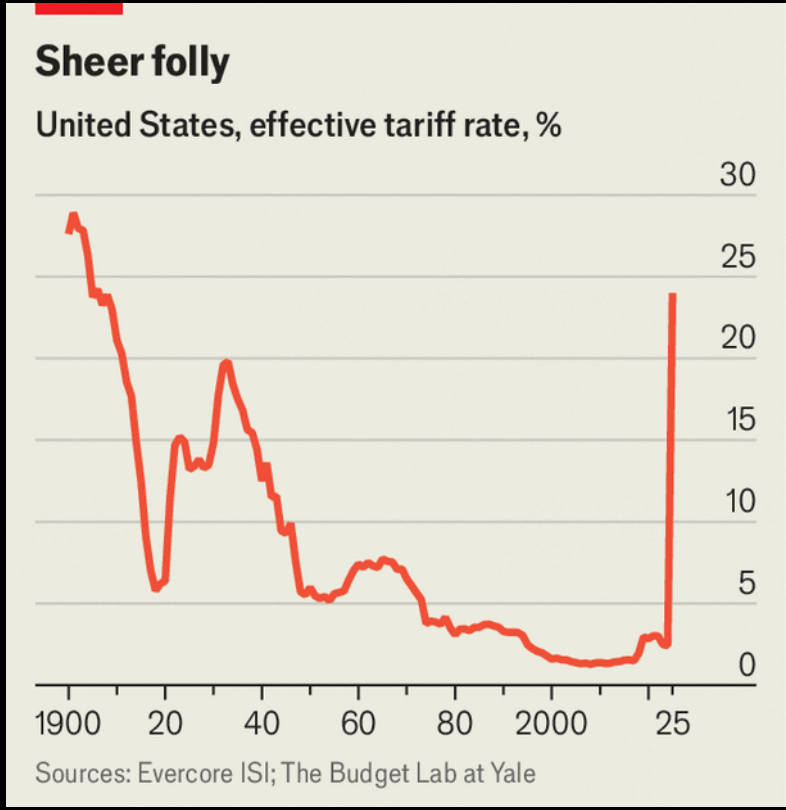


Why are tariffs bad?



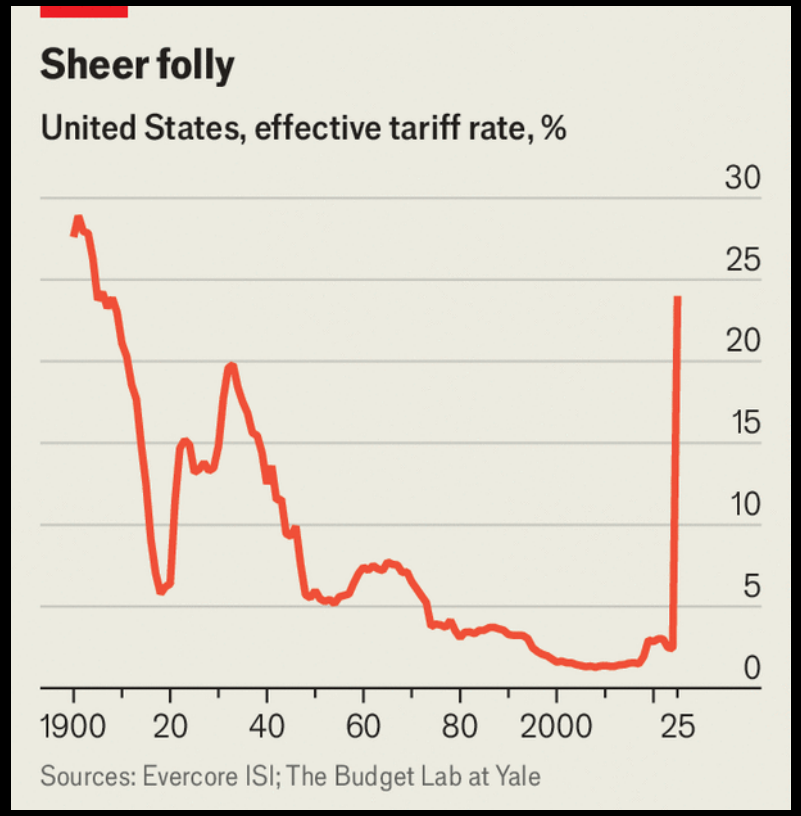
Why are tariffs bad?

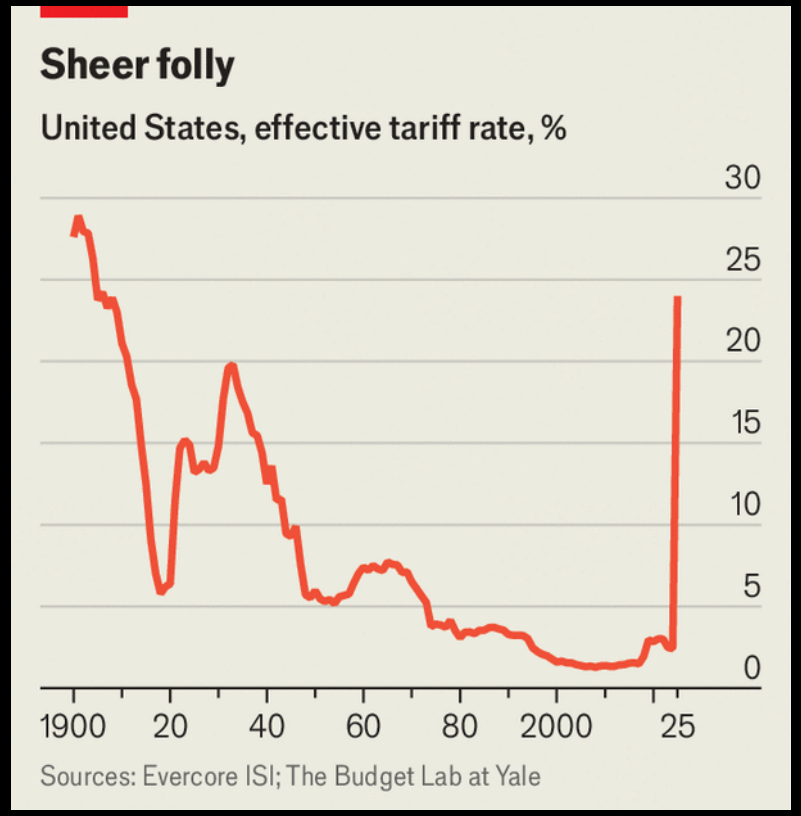
- One-off distortions (AA, CA). Static.



Why are tariffs bad?

- One-off distortions (AA, CA). Static.
- Block transfer of knowledge and ideas (HO model). Dynamic.



Why are tariffs bad?

- One-off distortions (AA, CA). Static.
- Block transfer of knowledge and ideas (HO model). Dynamic.
- Block specialization and cumulative scale agglomeration and learning (NT). Dynamic